P04: Makers Makin' It, Act II -- The Seequel

Design Doc by Da Real Big Cool, pd. 5

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PROJECT NAME: InfiniteIKEA
A Flask application backed by SQLite3 which allows users to dynamically compare IKEA items on an international scale. Points of comparison include statistics such as price & currency, or average review rating & # of reviews. The site will use D3.js for data visualization and Bootstrap for styling; React may be used for interactive components in data visualization.

TARGET SHIP DATE: 4/20/2026

Our MVP:

A website that provides geographic + statistical visualizations making it easier to compare IKEA products based on aforementioned parameters. A user system to save your favorite, interesting products. Informative and fun.

Program Components:

- ☐ Database: https://www.kaggle.com/datasets/ranjithpanicker/ikea-global-product-catalog-and-price-2026
  - ☐ To be preprocessed into SQLite format from csv for ease of use
  - ☐ Details below in database section
- ☐ sqlite3 (backend data storage system)
  - ☐ users (basic user account info, products saved)
  - ☐ products (product info used to compare on an international scale)
- ☐ Python (application layer)
  - ☐ __init__.py (runs Python, handles most backend data, database work etc)
- ☐ Flask (web server/delivery framework)
- ☐ html (frontend display)
  - ☐ register.html (adds new login info to database)
  - ☐ login.html (checks entered login info against database)
  - ☐ homepage.html (main dashboard, where the user can see top products + query different products)

- ☐ profile.html (user profile with bio & leaderboard & stats)
- ☐ edit_profile.html (edit profile bio and achievements)
- ☐ saved.html (shows all the products pinned by the user)
- ☐ <id>.html (specific page for an item, showing its details, visualizations, comparisons to versions of the same product from other countries)

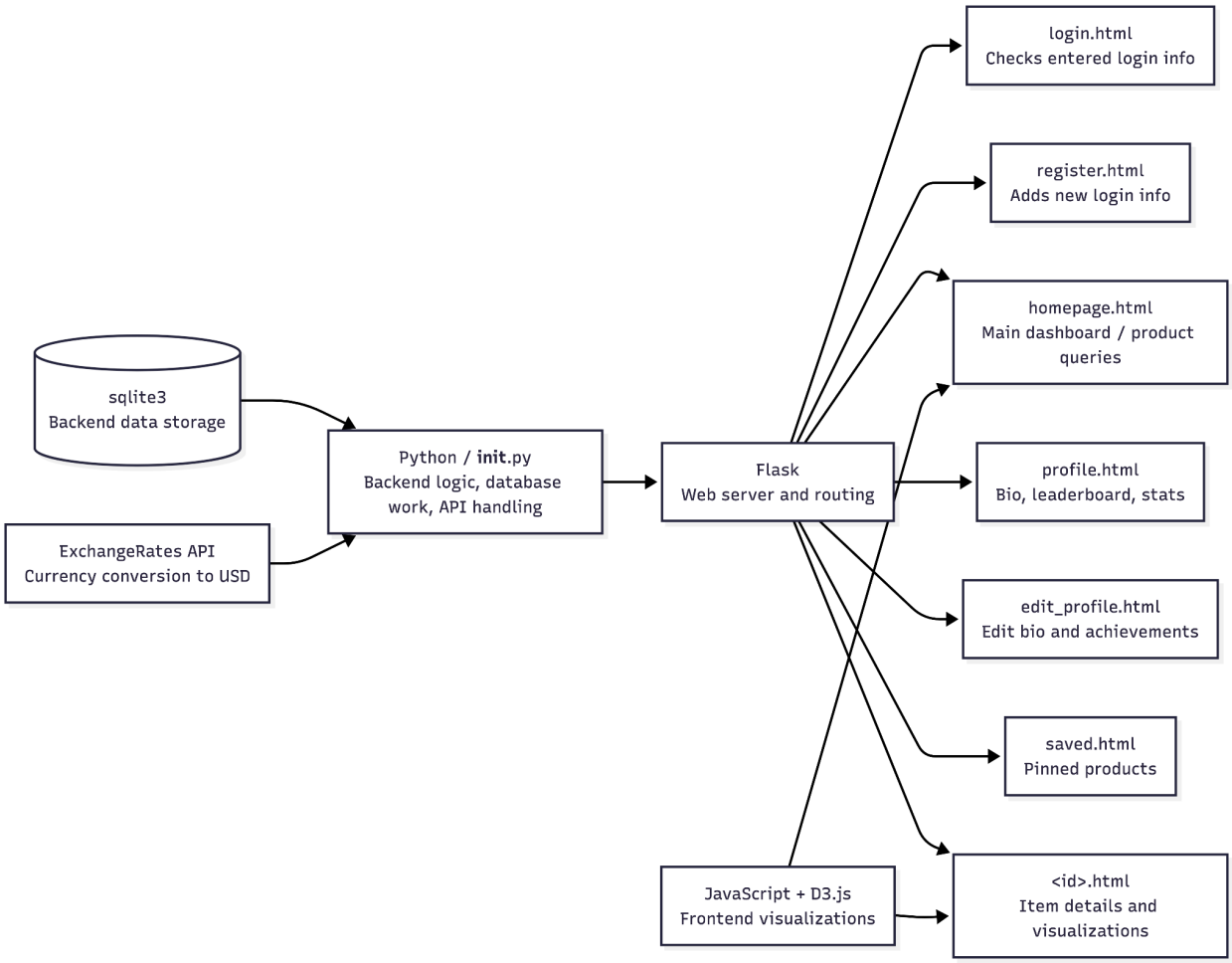
☐ Javascript (Frontend visualization)

- ☐ D3.JS (for data visualization)

☐ APIs

- ☐ https://www.exchangerate-api.com/ (to convert currencies to USD for comparison purposes)

Component Map:

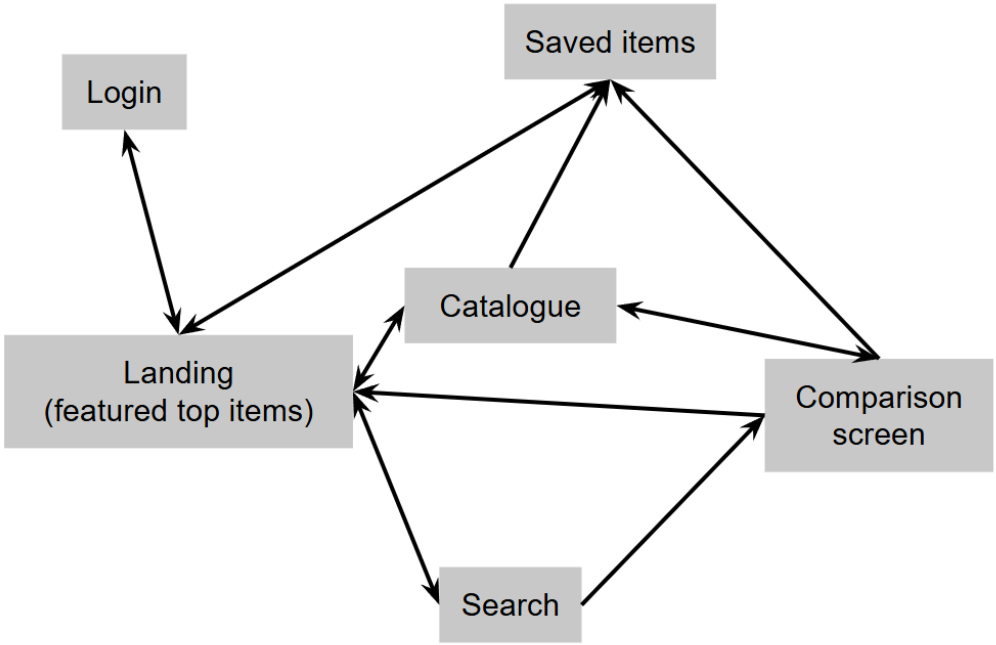


Database Organization:

users a master list of all registered users			
TEXT	username	PK	unique identifier of user
TEXT	email		
TEXT	password		
DATE	creation_date		only populated once during account creation
TEXT	saved_items		list of item ids user saved (can add to or remove from)

products a master list of all registered users			
TEXT	id	PK	unique identifier of each listing
TEXT	product_id		Used to identify identical items in different countries
TEXT	name		User-facing name
TEXT	currency		Three-letter code, so we can convert
FLOAT	price		Local currency price
TEXT	badge		Tracks popular/hot items, new, etc. tags provided by dataset
FLOAT	rating		Rating in that country
INT	rating_count		# of reviews contributing to rating

Site Map:



Front End Framework

We'll be using BootStrapCSS as our main front end framework to style our website. We prefer using BootStrap over other front-end frameworks because it is very versatile and many of our devos have experience with it, so it's best to work with what we have.

APIs

We will use the ExchangeRate API (https://www.exchangerate-api.com/) to convert local currency (chosen by user upon signing up) to a standardized one for ease of comparison within prices.

Data Visualization Library

We plan to use D3.js as the primary data visualization library. After comparing against simpler libraries, we chose D3.js as it is highly customizable and supports interactive/dynamic visualizations. Although the learning curve is steeper, its capabilities outweigh this drawback given our goals.

As our dataset details many products from different regions, we plan on comparing them statistically and displaying results geographically.

Planned Visualizations

- Grouped Bar Chart: Visualize selected items side-by-side. X-axis can represent countries or items, while the y-axis represents values such as price or rating.
- Sortable Bar Chart: Allows users to sort items based on selected criteria (e.g., ascending/descending price or rating).
- Choropleth Map: Provides a geographic view of the data with color intensity representing values such as price or number of reviews for a selected item across different countries.

Tasks & Assignments +

Task	Description	Responsible individual	Deadline	Status
Design Doc	Final	David	Mar 27, 2026 8:00 AM	Finished
App is up		Rohan	Mar 30, 2026	In progr…
Readme + install guide		Christine	Mar 31, 2026	In progr…
Database	Implement	David	Apr 13, 2026	Not begun

functions	functions to query/update the database			
API Integration	Connect to external API and process returned data	David	Apr 13, 2026	Not begun
Landing	Create homepage with featured items	Christine	Apr 13, 2026	Not begun
Search	Implement search functionality	Rohan	Date	Not begun
Catalogue	Display items list with relevant information (price, rating, etc.)	Christine	Date	Not begun
User system	Implement user accounts	Bogdan	Apr 13, 2026	Not begun
Comparison	Allow users to compare selected items across countries	Bogdan	Date	Not begun
Data Visualizations	Create graphs using D3.js	Christine	Date	Not begun
Frontend Styling	Design UI using Bootstrap & CSS	All	Date	Not begun